System Install

Pre-event setup - one list, everyone installs everything

Complete this setup well before the test begins, on reliable Wi-Fi. The total download is roughly 2 GB, so do not leave it for the last hour. At the end, one verify command must print OK - bring a laptop that already passes it. If it doesn't pass beforehand, fix it beforehand; "pip was still running" is not a timed-hour excuse.

Step 1 - Operating system

- Linux and macOS work as-is.
- **Windows: install WSL2 + Ubuntu first** (`wsl --install` in an admin PowerShell, then reboot). This is a multi-GB download - start it well in advance, not right before the test. Do everything below INSIDE Ubuntu/WSL.

Step 2 - Python 3.12

- Version 3.10–3.12 is acceptable; **3.13 will not work**. 3.12 is what we test with. Check: `python3 --version`
- **Apple-Silicon Mac users**: your Python must be native arm64. Check: `python3 -c "import platform; print(platform.machine())"` - it must print `arm64`. If it prints `x86_64`, install Python from python.org and use that one.
- **Intel Macs are not supported.** Bring a different laptop.

Step 3 - System tools (not pip)

- macOS:
```
xcode-select --install          # C compiler + make (~1.5 GB)
brew install espeak-ng
```
- Ubuntu / WSL:
```
sudo apt install build-essential espeak-ng
```
- Optional: if you prefer solving systems problems in Go or Rust instead of C/C++, also install that toolchain (go.dev/dl or rustup.rs).

Step 4 - Python packages (one environment, one command)

Make a virtual environment and keep it at a SHORT path - deep, cloud-synced folders break things:

```
cd ~ && mkdir speedrun && cd speedrun
python3 -m venv env
source env/bin/activate
```

Linux/WSL only, run this first (avoids a 2.5 GB CUDA download):

```
pip install "torch>=2.4,<3" --index-url https://download.pytorch.org/whl/cpu
```

Then **everyone** runs this exact command (pinned versions - do not "upgrade" anything):

```
pip install "setuptools<81" "torch>=2.4,<3" "numba==0.60.*" "llvmlite==0.43.*"
kokoro==0.9.4 resemblyzer==0.1.4 "soundfile>=0.12" numpy scipy scikit-learn
pandas librosa
https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl
```

This is the ~1–1.5 GB step. It covers every assignment.

Step 5 - Verify (must print OK)

```
python -c "import torch, numpy, scipy, sklearn, pandas, librosa, soundfile,
kokoro, en_core_web_sm; from resemblyzer import VoiceEncoder; print('OK',
torch.__version__)"
espeak-ng --version
cc --version && make --version
```

All three commands must succeed. Screenshot the output.

On the day

All data, model weights, and starter code are handed to you on a USB stick / zip - nothing model-related downloads during the session. Wi-Fi is available and AI coding assistants are allowed, but the schedule assumes your environment already verifies OK.